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Improving the Travelly App

A Human-Centered Approach to Promoting Active Transport in Southeast Queensland.

IFN637 Human-Centred Design of IT Systems

GROUP 27

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ACADEMIC HONESTY STATEMENT

I/We declare that all parts of this report are prepared by the below-listed members, and no uncited material from any external sources is included. I/We declare that this report genuinely reflects my/our activities and tasks performed as a part of assessments within the unit IFN637. The work contained in this report has not been previously submitted to any other units offered by QUT or any other universities around the world.

Furthermore I/We declare that (place a cross in each box):

- ☑ I/We did not cut-and-paste information from other people or AI-generated content without appropriate use of quotation marks and direct reference to their work;
- ☑ I/We did not re-word the ideas of others without proper and clear acknowledgement;
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- ☑ Any content generated with AI platforms such as ChatGPT is cited.

I/We understand that any violation of the above will result in a possible:

- Marks deduction for this assignment and/or
- The assignment will be reported to the Faculty Academic Misconduct Committee.

For assessment purposes, we give the assessor of this assignment the permission to: reproduce this assignment and provide a copy to another member of staff; and take steps to authenticate the assignment, including communicating a copy of this assignment to a checking service (which may retain a copy of the assignment on its database for future plagiarism checking).

Student Name	Number	Signature
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3. Introduction

The primary aim of this research project is to improve the design of the Travelly application using a human-centered design (HCD) process to better promote the use of walking, cycling, and public transport within Southeast Queensland. By collaborating with users in a real-world context, this study seeks to define concrete user needs and establish a clear problem-opportunity space. Specifically, the research aims to identify existing barriers to sustainable commuting behaviours and evaluate how the Travelly app can effectively influence and support positive behaviour change among urban commuters.

Transitioning from motorized transport to active modes provides significant holistic benefits. At an individual level, active commuting increases physical and mental well-being through daily activity. Literature supports this, demonstrating a clear correlation between active modes of transportation and increased physical activity across the life course (Prince et al., 2022). On a macro level, decreasing reliance on motorized transport reduces traffic congestion and the region's overall environmental carbon footprint. Furthermore, platforms like Travelly serve an important infrastructural role; the app collects vital travel data that can be utilized by the Department of Transport and Main Roads (TMR) to inform future transport planning and development.

To ensure the proposed solution is genuinely user-centric, the project is grounded in participatory design approaches (Sanders, 2002) that prioritize deep user engagement. The theoretical evaluation of the current Travelly prototype is framed around Peter Morville's User Experience Honeycomb (Morville, 2004). This framework specifically allows the research team to evaluate the platform's utility (e.g., assessing if it effectively helps people find safe bike paths) and its credibility (e.g., assessing whether users trust TMR to handle their personal travel data securely).

4. Methods

4.1. Research Aim and Methodological Approach

The aim of the research methodology is to collaborate with end-users to understand their experiences and gather actionable data to define the design problem. To achieve comprehensive insights, the research employs a triangulation strategy integrating "Say," "Make," and "Do" methods. This combined framework allows the research team to document what people explicitly state they do, observe their actual behaviours in simulated scenarios, and externalize their deeper, unspoken aspirations for better travel.

4.2. Participants and Recruitment

- The target demographic for this study is urban commuters in Southeast Queensland who are open to utilizing sustainable transport alternatives.

- Participants selected were adults aged 18 to 65 who live or work in Brisbane.
- Additional criteria required participants to be frequent technology users who are motivated by either physical fitness or cost-saving benefits.
- Recruitment was conducted via direct outreach using a tailored email and message that clearly explained the study's aim.
- Participants were identified and recruited from networks of students, colleagues, local community groups, and university bulletin boards.
- To meet the study's scope, at least one participant was recruited per group member, ensuring a sufficient sample size for qualitative analysis.

4.3. Ethical Conduct

- All research activities were planned and executed in alignment with the National Statement on Ethical Conduct in Human Research to ensure transparency and ethical practice (National Health and Medical Research Council et al., 2025).
- Signed consent forms were collected from all participants prior to the commencement of any primary research.
- To protect participant confidentiality, all study data and interview recordings were transcribed and anonymized by changing real names to pseudonyms.
- Any visual data collected was by participant's explicit confirmation, before recordings participants were notified well in advance.
- All ethics documentation, including the recruitment email, information sheet, and tailored consent form template, are provided in Appendix A.

4.4. Data Collection Methods (Reordered to Say, Make, Do)

The investigation utilized three distinct user research instruments, the details of which are cross-referenced in Appendix B:

- **Say Method (Semi-Structured Interview):** To establish baseline behaviours, participants were asked to walk researchers through their typical commutes. The interviews focused on identifying the biggest challenges users face when trying to cycle or use public transport, alongside exploring the core motivations that would encourage them to choose walking or cycling over driving for short trips.
- **Make Method (Experience Mapping):** To uncover latent needs, participants engaged in a generative mapping exercise. Participants were given a digital whiteboard with tools and asked to draw a map of their daily experience of journey.

They were instructed to visually highlight specific "Pain Points" (e.g., steep hills or a lack of shade) and "Delight Points" (e.g., safe crossings or scenic views). The goal of this process was to externalize beliefs and perceptions about the local environment that may not surface in a standard verbal interview setting.

- **Do Method (Scenario-Based Usability Test):** This method observed actual user interactions within a transit app environment. Because the Travelly app prototype experienced technical issues during the research phase, participants were tasked with using a functional proxy (such as Google Maps or TransLink). Participants were provided with a scenario where they needed to commute along frequently used routes. They were tasked with using the proxy app to find the safest route and record their trip. Utilizing the "Thinking Aloud" technique, participants verbalized their thought processes, decisions, and frustrations while navigating the interface.

4.5. Data Analysis and Limitations

Research sessions were documented through audio/video recordings, supplemented by researcher notes and transcriptions. Due to technical challenges, few transcripts failed to generate natively, so the audio recordings were subsequently transcribed with the assistance of an online tool to preserve privacy. Links to the secure recordings are available in Appendix D. The qualitative data collected from all methods was then thematically analysed against the Human Centred Design theoretical framework to extract key themes and concrete evidence of the user experience. Evidence of the anonymized study data and the subsequent analysis process is provided in Appendix C.

While the triangulation approach yields rich qualitative data, certain limitations must be acknowledged. The sample size is relatively small and heavily reliant on convenience sampling from immediate community and university networks, which may introduce self-selection bias. Additionally, while the scenario-based usability test effectively gauges navigational utility, it occurs in a simulated environment and cannot fully replicate the physical and environmental variables of an active commute. Furthermore, because the usability testing was conducted using Google Maps and TransLink as functional proxies for the unavailable Travelly prototype, the observed interactions reflect general route-planning behaviors rather than direct usability metrics of the Travelly interface itself.

5. Findings

The findings presented in this section are the result of a rigorous thematic analysis conducted on the qualitative data gathered during the primary research phase. By triangulating data from generative interviews (Say), usability observations (Do), and participatory journey mapping (Make), the research team was able to move beyond surface-

level complaints to uncover the underlying behavioral drivers and pain points of Southeast Queensland commuters. All participant data has been anonymized in accordance with ethical research guidelines. Four distinct, overarching themes emerged from this synthesis.

5.1. Unpredictability and Inaccuracy of Real-Time Tracking

The most universally cited frustration among all participants was the acute unreliability of real-time transit data across existing platforms (such as Google Maps and the TransLink app). Participants frequently noted a severe disconnect between the scheduled timetables and the physical arrival of vehicles. Participant explicitly noted during a generative interview, *"timing isn't fully accurate, especially when there are delays or traffic changes. Real-time updates could be... more reliable."* This issue was further corroborated during the usability testing (Do method). Participants demonstrated how they cross-reference multiple apps when a bus fails to arrive, highlighting a phenomenon colloquially known as the "ghost bus" where a vehicle appears on the app's GPS interface but abruptly disappears without physically arriving at the stop. Participant pointed out that during peak hours, Apps often inaccurately displays multiple buses merging or arriving simultaneously. This technological unreliability fundamentally erodes user trust. When faced with this unpredictability, particularly in the morning when time sensitivity is highest, users reported a high likelihood of abandoning public transport entirely in favor of driving, simply to regain a sense of control over their commute.

5.2. Overcrowding and Capacity Uncertainty

The shift toward hybrid working models has inadvertently created hyper-concentrated peak commuting days, fundamentally altering the user experience of public transit. Participants emphasized that mandatory office attendance policies frequently enforced on certain days like Tuesdays or Thursdays lead to severe and unpredictable overcrowding.

During a participatory mapping exercise (Make method), participant mapped their emotional journey, revealing a stark contrast between a "regular" commute and a "mandatory office day" commute. While early or off-peak buses yield a positive emotional state, the necessity to stand on a fully packed bus for 40 minutes on a Tuesday drastically reduces user satisfaction and leads to feelings of disappointment and physical discomfort. The research identified a critical gap: current applications offer route planning but completely lack predictive capacity indicators. Highlighting this need, participant creatively suggested: *"If the app could show how vacant the bus is, like a battery percentage display... maybe we can avoid the bus altogether instead of getting it to stop and then not getting a seat."* This indicates a strong user desire for agency and informed decision-making regarding physical comfort.

5.3. Infrastructure and Weather Dependencies

While the project initially hypothesized that users lacked the motivation to engage in active transport (walking or cycling), the findings revealed that motivation is highly dependent on external environmental and infrastructural factors. Participant who frequently commutes via walking, emphasized that physical infrastructure dictates their transport choices. They noted, *"Infrastructure is the key factor. If there is better infrastructure to support walking (shade, comfortable environment), [I] would walk more."*

Furthermore, sudden adverse weather events, coupled with inadequate infrastructure at transit stops, act as a primary deterrent to sustainable travel. Participants expressed deep frustration over arriving at a bus stop only to find no shelter during sudden rainstorms, resulting in a physically uncomfortable journey before they even reach the office. The data suggests that for a transit app to successfully encourage active or public transport, it must integrate hyper-local weather forecasting and infrastructure-aware routing (e.g., highlighting shaded walking paths or sheltered stops).

5.4. Pragmatic Convenience over Environmental Altruism

A core objective of this research was to understand how to promote eco-friendly travel behaviours. However, the findings explicitly challenge the assumption that users will choose sustainable options purely for environmental benefits. Participants universally prioritized pragmatic factors namely time, convenience, and financial cost over ecological impact.

When queried about whether environmental impact influences their daily travel choices, a participant stated definitively: *"I do think about it, but it doesn't always influence my daily decisions. Most of the time I prioritize time and convenience, especially with my early work schedule."* However, while environmental altruism failed as a primary motivator, financial incentives proved highly effective. Multiple participants organically praised the recently introduced 50-cent public transport fares in Queensland, citing it as a major catalyst for choosing the bus over driving. Therefore, the findings conclude that any digital intervention aiming to promote sustainable transport must frame those choices through the lens of user convenience, financial savings, and time efficiency.

6. Design specifications

To effectively translate the complex, multi-faceted research findings into actionable frameworks for the ideation and prototyping phases, the team developed two complementary design specifications. In alignment with established Human-Centered Design methodologies, we selected

- **Persona** (to encapsulate the demographic, psychographic, and technological profile of our target audience)
- **User Journey Map** (to visualize the chronological sequence of activities, pain points, and emotional fluctuations in context).

Together, these artifacts provide a robust, empathetic foundation for the subsequent design brief.

6.1. Primary Persona: "Ian" (The Pragmatic Hybrid Commuter)

The "Ian" persona is a composite archetype synthesized directly from the generative interviews and usability tests, representing the core behaviours and frustrations of the young professional commuter demographic in Southeast Queensland.

High Quality implementation of Persona can be accessed through link below as well

https://miro.com/app/board/uXjVHZIk3Vk=/?share_link_id=266146795011

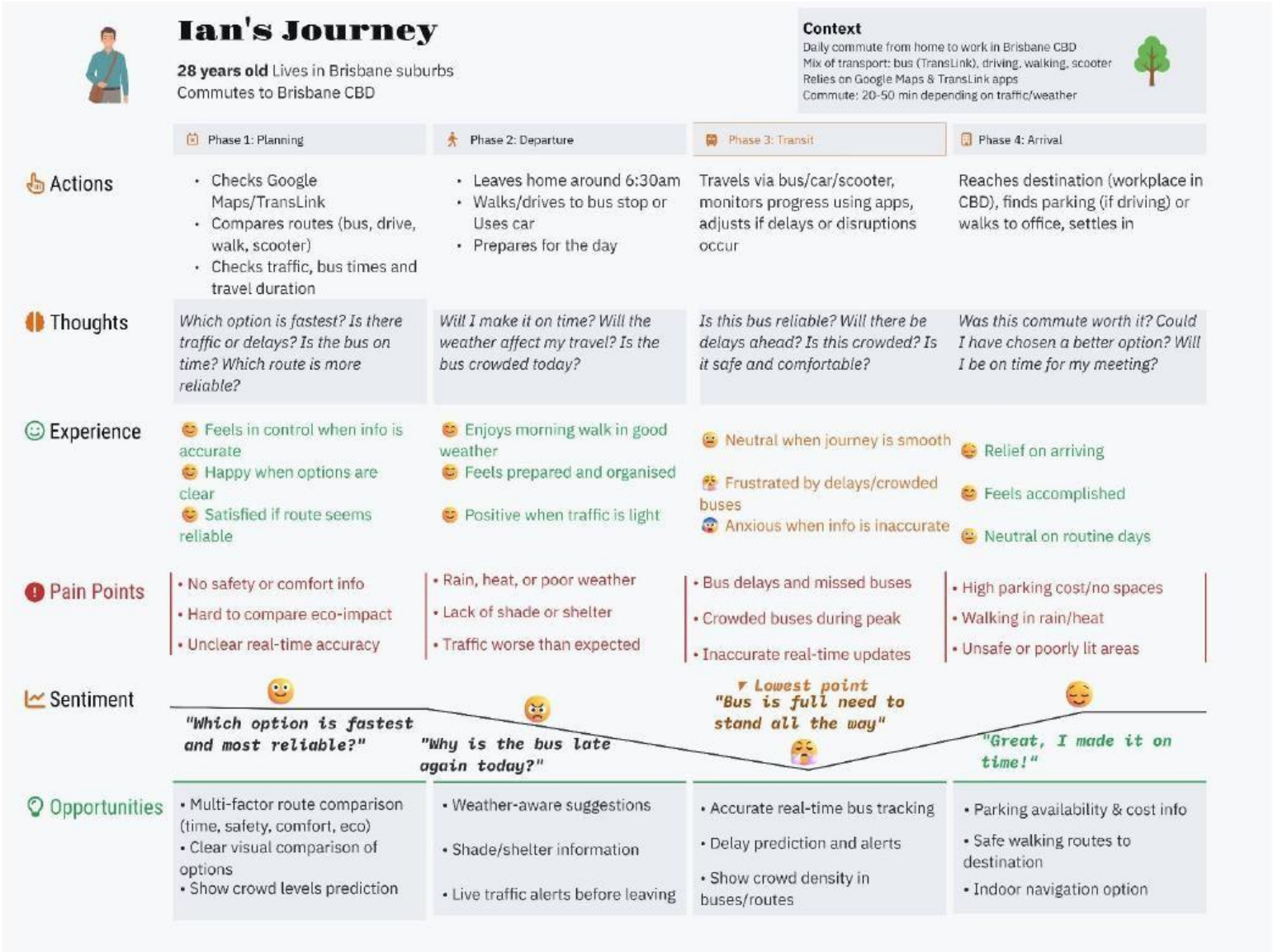


6.2. User Journey Map: The Morning Commute

This journey map specifically isolates the highest-friction scenario identified in the research: the mandatory in-office commute on a Tuesday. By mapping this specific context, the team can pinpoint exactly where digital interventions will have the highest impact.

High Quality implementation of Journey map can be accessed through link below as well

https://miro.com/app/board/uXjVHZIk3Vk=/?share_link_id=266146795011



7. Design brief

The following design brief serves as a strategic roadmap, translating the synthesized research and design specifications into defined parameters for the upcoming ideation and prototyping phases of the Travelly application.

7.1. Problem Space to be Addressed

Commuters in Southeast Queensland are experiencing high levels of anxiety and frustration due to the unpredictability of the public transport network. This friction is driven by inaccurate real-time vehicle tracking ("ghost buses") and severe, unpredictable overcrowding exacerbated by post-pandemic hybrid working schedules. This uncertainty, compounded by poor localized infrastructure (such as a lack of weather shelters at suburban stops), frequently drives users to abandon sustainable transport options in favour of the perceived control and comfort of private vehicles.

7.2. Objectives and Goals of the New Design

To successfully pivot user behaviours toward sustainable commuting, the redesigned Travelly app must prioritize reliability, transparency, and user agency. The core objectives are framed by the following "How Might We" (HMW) statements:

- *How might we* provide commuters with highly accurate, real-time visibility into both transit vehicle locations and predictive crowd-density, allowing them to make informed decisions before leaving their homes?
- *How might we* integrate weather-aware and infrastructure-aware routing (e.g., mapping shaded walking paths or sheltered transit stops) into the journey planning experience to reduce environmental deterrents?
- *How might we* highlight pragmatic benefits such as the 50-cent fare cost savings and precise time-to-destination metrics to make public transit the undeniable path of least resistance?

7.3. Target Audience

The primary target audience comprises young professionals and tertiary students (encapsulated by the "Ian" persona) living in the middle-ring suburbs of Southeast Queensland who commute regularly to central business and educational hubs (e.g., Brisbane CBD, South Bank). These users are highly tech-savvy, time-poor, and value seamless digital experiences. Crucially, while they are open to sustainable and active travel, their choices are ultimately governed by convenience, reliability, and personal comfort rather than environmental altruism.

7.4. Project Scope

- **In Scope (Minimum Viable Product):** The development of a high-fidelity mobile app prototype focused on the pre-planning and transit-waiting phases. Key features will include an interface for real-time GPS tracking of buses/trains, a predictive crowd-level indicator (utilizing a visual capacity percentage or traffic-light system), and integrated weather alerts tied to specific user-saved routes.
- **Out of Scope:** The physical construction or funding of municipal infrastructure (e.g., building bus shelters or planting shade trees); the backend engineering of the actual GPS tracking hardware; and the inclusion of gamification features focused solely on tracking carbon footprints, as the primary research clearly indicated this does not motivate the target demographic.

8. GenAI reflection

Overview

This section provides an in-depth reflective account of our team's integration of Generative AI tools (such as ChatGPT/Gemini) during the research and synthesis phases of Assignment 2. The aim of this assessment was to conduct research with people in a real-world context and define their needs to share with an industry partner. As such, ensuring the accuracy and professionalism of our findings was paramount. This reflection is structured using Driscoll's (1994) framework *What?, So What?, and Now What?* to explore the impact of GenAI on our human-centred design workflow, the ethical challenges we encountered, and our strategies for maintaining research integrity.

1. WHAT? (Description of the Event)

In this assignment, our team utilized Generative AI not as a primary researcher, but as a supplementary tool to overcome specific logistical hurdles, structural challenges, and drafting bottlenecks. We applied it across four distinct areas of our workflow:

- **Transcription Fallback and Recovery:** During our primary data collection, we initially relied on Zoom's native transcription feature. However, due to technical errors, Zoom failed to generate transcripts for specific segments of our audio recordings. To recover this data, we utilized a Generative AI audio-to-text tool to process the raw, localized audio files and generate the missing transcripts, allowing us to proceed with our analysis without losing valuable participant insights.

- **Thematic Validation:** After our team manually coded the transcripts and derived our initial themes based on user behaviours and pain points, we used GenAI as a secondary validation tool. We inputted the anonymized transcripts and prompted the AI to identify recurring patterns. We then cross-referenced these AI-generated themes against our human-derived themes to ensure our manual synthesis had not overlooked any subtle, underlying patterns in the qualitative data.
- **Design Specification Brainstorming:** Moving into the creation of our design brief, we needed to format our findings clearly for the industry partner. We used GenAI to brainstorm structural templates for our Design Specifications (specifically Personas, Journey Maps, and Scenarios). We used prompts such as, *"Provide a comprehensive structural layout for a UX Journey Map tailored to [Insert specific user context]."*
- **Refining and Formalizing Tone:** Finally, translating raw user quotes and rough thematic notes into a cohesive, professional report can be time-consuming. We utilized GenAI to assist in rewriting sections of our draft. We inputted our rough, human-written paragraphs and prompted the AI to "rewrite this text in a formal, academic, and professional tone suitable for an industry-facing UX research report," ensuring our final submission was polished and coherent.

2. SO WHAT? (Analysis and Ethical Considerations)

The integration of GenAI significantly accelerated our administrative tasks and enhanced the professional presentation of our report. However, it also introduced severe quality-control challenges that required active, continuous management. We evaluated these challenges through the lens of four core ethical values: respect, research merit and integrity, justice, and beneficence.

- **Respect and Privacy:** The foremost ethical hurdle was handling raw user data during the transcription and thematic validation phases. To uphold the value of respect, we implemented a strict anonymization protocol. Before any audio file or text was processed by a third-party GenAI model, all Personally Identifiable Information (PII) including names, locations, and identifying workplace details was scrubbed from the dataset.
- **Research Merit and Integrity (Combating Hallucinations):** This was the most critical challenge we faced. While using AI for thematic validation and tone-rewriting was useful, the AI was highly prone to "hallucinations." In several instances, the AI outputted confident statements, fabricated participant quotes, or exaggerated user pain points to make a more compelling narrative. Furthermore, when asking the AI to

rewrite text into a formal tone, it sometimes subtly altered the *meaning* of the original insights.

- **Our Mitigation Strategy:** To maintain research integrity, we established a strict "trust but verify" rule. Absolutely no AI-generated content was accepted at face value. We conducted line-by-line reviews of the formalized text and systematically removed any AI hallucinations. Every theme, quote, and synthesized point in our final report was manually traced back to the original, human-verified transcripts to ensure our findings were authentically grounded in empirical data.
- **Justice:** When brainstorming templates for Personas, Journey maps and Scenarios, we had to be highly cautious of the AI introducing demographic or behavioural biases. GenAI models often default to generalized stereotypes when generating user personas. To ensure justice and fair representation, we used the AI strictly for the *structural layout* (e.g., categories, headers, mapping stages). The actual characteristics, pain points, and behaviours populated into those templates were derived exclusively from our unbiased user research.
- **Beneficence:** Ultimately, using AI to recover lost transcripts, brainstorm structures, and polish our professional tone maximized the efficiency of our report. By offloading these tasks to the AI, our team could redirect our cognitive efforts toward what truly matters in Human-Centred Design: deep empathetic analysis and generating high-quality, actionable insights that will practically benefit the users and the industry partner.

3. NOW WHAT? (Lessons Learned and Future Action)

This experience fundamentally shifted my perspective on the role of Generative AI in the UX design process.

- **Key Lessons Learned:** The primary takeaway is that GenAI is a powerful "co-pilot" for structural organization, tone refinement, and administrative recovery, but it is a highly flawed substitute for human empathy and critical analysis. The frequent presence of hallucinations proved that AI cannot be trusted to independently synthesize qualitative human experiences. It tends to prioritize a "good sounding story" over factual accuracy. Furthermore, I learned that the time saved by having AI generate content is often offset by the meticulous time required to rigorously fact-check that same content against the raw data.
- **Future Application:** Moving forward into next steps (which focuses on ideation and sketching), We will continue to utilize GenAI to overcome "blank page syndrome,"

brainstorm diverse design concepts, and polish written rationale. However, I will implement an even stricter validation checklist. I will treat all AI outputs not as definitive answers, but as raw, potentially flawed material that must be subjected to rigorous human scrutiny and ethical review before being integrated into any final design solution. Human intuition and deep engagement with the actual user data must always remain the final arbiter in the design process.

9. References

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10. Appendix

10.1 Appendix A: Ethics Documents

10.1.1 Consent form

CONSENT FORM FOR QUT RESEARCH PROJECT
Interview & Experience Mapping & Usability Testing

Improving the Travelly App

IFN637 Human-Centred Design of IT Systems
QUT Ethics Approval Number 9541

Research team contacts

Student/s:	Name/s:	QUTEmail/s:
1	Dileep Samala	[redacted]
2	Manoj Godavarthi	[redacted]
3	Srinivasa Teja Bharadwaj Penumudi	[redacted]
4	Lokesh Kunapaneni	[redacted]
Unit Coordinator:	A/Prof Bernd Ploderer	07 3138 4927

Faculty of Science, Queensland University of Technology (QUT)

Statement of consent

By signing below, you are indicating that you:

- Have read and understood the information document regarding this research project.
- Have had any questions answered to your satisfaction.
- Understand that if you have any additional questions, you can contact the research team.
- Understand that you are free to withdraw without comment or penalty.
- Understand that if you have concerns about the ethical conduct of the research project you can contact the Research Ethics Advisory Team on 07 3138 5123 or email [redacted]
- Understand that the research project will include an audio, screen, and/or video recording.
- Understand that non-identifiable data from this project may be used by QUT or a partner organisation to improve their digital products and services or as teaching material.
- Agree to participate in the research project.

Name: _____

Signature: _____

Date: _____

Please return this signed consent form to the researcher.

10.1.2 Participant Information Sheet

PARTICIPANT INFORMATION FOR QUT RESEARCH PROJECT
– Interview & Usability Testing & Experience Mapping

Improving the Travelly App: A Human-Centered Approach to Promoting Active Transport in Southeast Queensland

IFN637 Human-Centred Design of IT Systems

QUT Ethics Approval Number 9541

Research team

Student/s:	Name/s:	QUTEmail/s:
1	Dileep Samala	[redacted]
2	Manoj Godavarthi	[redacted]
3	Srinivasa Teja Bharadwaj Penumudi	[redacted]
4	Lokesh Kunapaneni	[redacted]

Unit Coordinator: A/Prof Bernd Ploderer [redacted] 07 3138 4927

Faculty of Science, Queensland University of Technology (QUT)

Why is the study being conducted?

This research project is being conducted as a component of the master's program for students enrolled in IFN637 Human-Centred Design of IT Systems.

The primary objective of this research is to examine how urban commuters in Southeast Queensland utilize and engage with the Travelly app. Furthermore, the study aims to identify potential enhancements to its design that would more effectively encourage walking, cycling, and the use of public transport. This research is being conducted in collaboration with the Travelly app's development team.

We are seeking your participation as you are an urban commuter who regularly employs various modes of transportation and may utilize digital tools for the purpose of trip planning.

What does participation involve?

Your participation will involve the following activities:

1. Interview (Say Method)
   This will be a semi-structured interview lasting approximately 20–30 minutes and will be audio-recorded. Sample questions include:
   i. Can you describe your typical daily commute?
   ii. What challenges do you face when using public transportation or cycling?
   iii. What would encourage you to choose active transportation over driving?

2. Usability Test (Do Method)
   You will take part in a usability and user experience evaluation of the Travelly app prototype, lasting approximately 30–40 minutes. During this session, you will be asked to complete a scenario such as:
   Finding a route using either cycling or public transportation within the app.
   Please note:
   i. Your interactions may be screen recorded.
   ii. You may be asked to verbalize your thoughts while using the app.
   iii. Observations will be made to identify any usability issues.

3. Experience Mapping (Make Method)
   This activity will take approximately 20 minutes. You will be asked to:
   i. Draw or describe your daily travel journey.
   ii. Highlight any challenges (pain points) and positive experiences (delight points) you encounter along the way.

What are the possible benefits for me if I take part?

It is expected that this research project will not provide direct personal benefits to participants. However, your participation will help improve the design and usability of the Travelly app, which aims to promote more sustainable and efficient travel options, such as walking, cycling, and public transport, in Southeast Queensland.

Additionally, your feedback will help IFN637 students gain a deeper understanding of real user needs and apply human-centred design methods in a practical context, thereby enhancing their professional skills. The insights gathered may also help the Travelly app development team refine their digital services to better meet user expectations.

What are the possible risks for me if I take part?

There are some risks associated with your participation in this research project. These include potential inconvenience from the time required to participate, possible discomfort from being audio- or screen-recorded during the interview and during usability testing, and a minor risk of being identified due to the small number of participants.

To minimise inconvenience, all research activities will be carefully planned to ensure they are relevant and efficient. To reduce discomfort from being recorded, sessions will be conducted in a comfortable, private environment, and participants will be informed in advance of any recordings. To minimise the risk of identification, all data will be anonymised using pseudonyms, and any identifying details will be removed before analysis or reporting.

QUT provides for limited free psychology, family therapy or counselling services for research participants of QUT research projects who may experience discomfort or distress as a result of their participation in the research. Should you wish to access this service, please call the Clinic Receptionist on **07 3138 0999** (Monday–Friday only 9 am–5 pm), QUT Psychology and Counselling Clinic, 44 Musk Avenue, Kelvin Grove, and indicate that you are a research participant. Alternatively, Lifeline provides access to online, phone or face-to-face support; call **13 11 14** for 24-hour telephone crisis support. If you are aged up to 25, you can also call the Kids Helpline on **1800 551 800**.

What about privacy and confidentiality?

All comments and responses are coded i.e. it will be possible to re-identify you. A re-identifying code stored separately to personal information (e.g. name, address), will only be accessible to the research team, and the code plus identifying information will be destroyed at the end of the semester.

The information that will be removed includes names, place of work/study, occupation, education, travel behavior (e.g., commute patterns and transport preferences), general location information (e.g., frequently visited areas), and app usage behavior collected during interviews and usability testing

Any data collected as part of this research project will be stored securely as per QUT's Management of research data policy. Data will be stored for a minimum of 5 years and can be disclosed if it is to protect you or others from harm, if specifically required by law, or if a regulatory or monitoring body such as the ethics committee requests it.

As the research project involves an audio, screen or video recording:

- You will not have the opportunity to verify your comments and responses prior to final inclusion.
- The recording will be retained for the minimum retention period of 5 years after the last research activity.
- The recording may be used for teaching purposes and to support improvements to the Travelly app and its associated digital services.
- Only the named researchers will have access to the original recording.
- It is not possible to participate in the research project without being recorded.

Every effort will be made to ensure that the data you provide cannot be traced back to you in reports, publications and other forms of presentation. For example, we will only include the relevant part of a quote, we will not use any names, or names will be changed, and/or details such

as dates and specific circumstances will be excluded. Nevertheless, while unlikely, it is possible that if you are quoted directly your identity may become known.

How do I give my consent to participate?

We would like to ask you to sign a written consent form (enclosed) to confirm your agreement to participate.

What if I have questions about the research project?

Please contact the researchers (details above) to have any questions answered or if you require further information about the project.

What if I have a concern or complaint regarding the conduct of the research project?

QUT is committed to research integrity and the ethical conduct of research projects. If you wish to discuss the study with someone not directly involved, particularly in relation to matters concerning policies, information or complaints about the conduct of the study or your rights as a participant, you may contact the QUT Research Ethics Advisory Team on 07 3138 5123 or email [redacted]

Thank you for helping with this research project. Please keep this sheet for your information.

10.1.3 Recruitment email

Dear [Participant],

My name is [Student name] and I am a postgraduate student at Queensland University of Technology (QUT) doing a research project into the experience with Travelly app.

I would like to invite people who travel daily for their work and use public transport daily to take part in research.

Please view the attached Information Sheet and Consent Form for further details on the study.

Please note that this study has been approved by the QUT Human Research Ethics Committee (approval number 9541).

Many thanks for considering this request.

If you are interested, could you please provide me the signed consent form.

Please let me know, if you have any questions

Thanks,

[Student name]
Post graduate Student
[Student email]

Associate Professor Bernd Ploderer
Unit Coordinator
07 3138 4927
[redacted]
School of Computer Science, Faculty of Science
Queensland University of Technology

10.2 Appendix B: Methods Used

10.2.1 Say method Semi-structured interview questions

1. Warm-up & Context

1. **Can you tell me about your current study or work routine?**
   Probe: Typical hours, flexibility, location
2. **Walk me through a typical weekday commute from leaving home to arriving at your destination.**
   Probe: Time taken, transport modes, routes

2. Travel Behaviour & Decision-Making

3. **How do you usually decide which transport option to take each day?**
   Probe: Time, cost, convenience, weather, mood
4. **What transport options do you use most often, and why those?**
   Probe: Habits vs conscious choice
5. **When was the last time you changed your usual travel method? What caused that change?**
   Probe: Disruptions, experimentation

3. Pain Points & Barriers

6. **What challenges do you face during your daily travel?**
   Probe: Delays, cost, comfort, reliability
7. **Are there situations where walking, cycling, or using public transport feels difficult or impractical?**
   Probe: Infrastructure, distance, safety

8. **Do you ever feel unsafe while travelling? If yes, in what situations?**
   Probe: Time of day, location, transport type

9. **What usually stops you from choosing more active or eco-friendly transport options?**
   Probe: Time pressure, effort, motivation

4. Current Solutions & Apps

10. **What apps or tools do you currently use to plan or manage your travel?**
    Probe: Google Maps, TransLink, etc.

11. **What do you like and dislike about these tools?**
    Probe: Accuracy, usability, trust, missing features

5. Attitudes & Motivation

12. **How important is environmental impact when choosing how you travel?**
    Probe: Does it influence decisions or not?

13. **What would make you more likely to walk, cycle, or use public/shared transport more often?**
    Probe: Convenience, incentives, infrastructure

6. Needs & Opportunities

14. **If something could improve your daily travel experience, what would it be?**
    Probe: Open-ended, let them lead

15. **What would you want to improve about your current travel experience?**
    Probe: Focus on outcomes, not features

10.2.2 Make method Experience mapping

Purpose

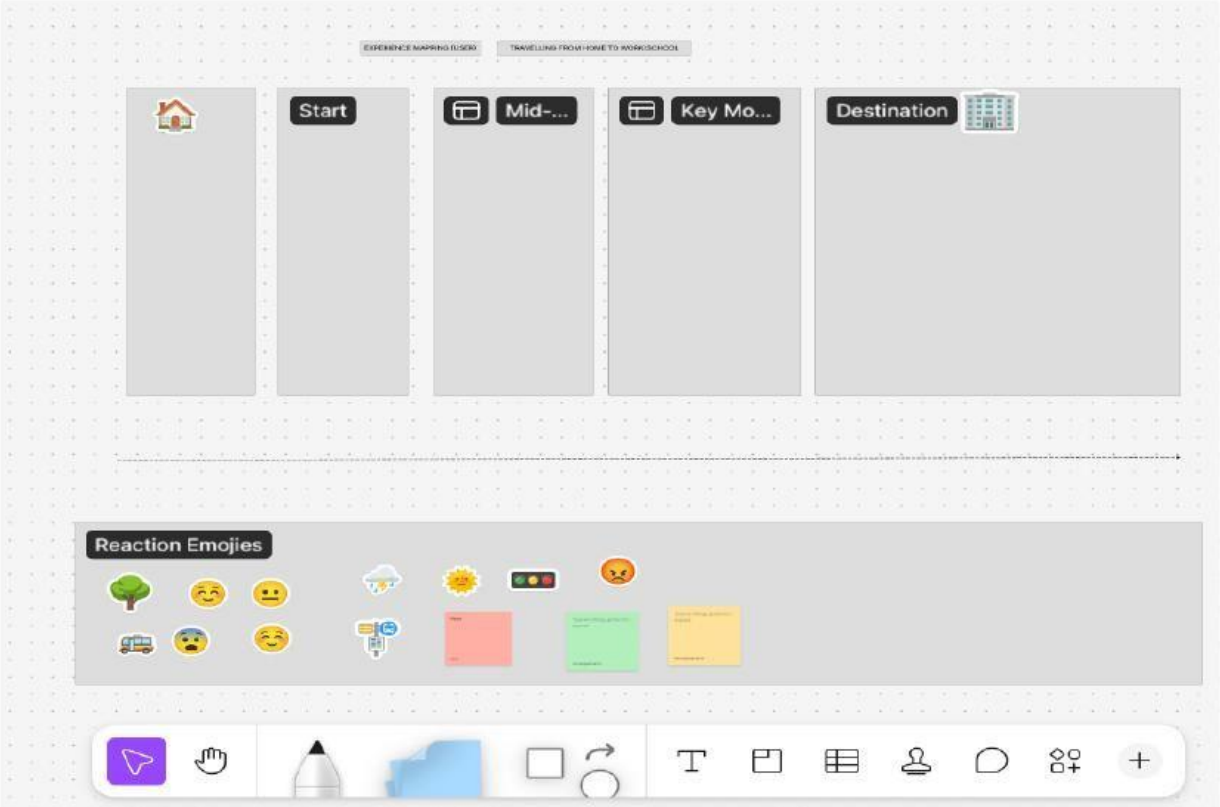
This activity aims to help participants **visually express their daily travel experience**, identifying both challenges and positive moments that may not emerge through interviews

Materials (Instruments)

Participants will be provided with:

The research employs a Fig Jam digital whiteboard as a sentiment-mapping tool. Participants react to each phase of their journey with emojis as well as sticky notes transforming their lived experience into a visual data set that highlights the emotional impact of waiting, transit, and arrival.

Instructions for Participants



1. Think about your **typical daily commute or travel journey**.
2. Along the journey, mark:
    a. **Unpleasant experiences**
    (e.g., steep hills, unsafe crossings, lack of shade, traffic issues)
    b. **Pleasant experiences**
    (e.g., scenic routes, safe paths, shaded areas, enjoyable spots)
3. You may add notes, symbols, or drawings to explain your experience.
4. Take your time and feel free to express your thoughts creatively.

What We Observe (take notes)
Pain points
Delight points
Emotional responses

10.2.3 Do method Usability Testing

Setup

Open **Travelly app prototype (We used Google Maps since Travelly app is not working and for participant convenience)**

Start **Zoom recording (with consent)**

Prepare:

- Consent form

- Observation notes sheet

Script

Introduction

My name [Student name] is and thank you for participating today.

1. You've been invited to help us understand **how well the Travelly app supports your commuting experience**, and where it can be improved.
2. Even though this is called a usability test, **you are not being tested**, the app is. There are no right or wrong answers.
3. Your honest feedback will help us improve how Travelly supports **walking, cycling, and public transport choices**.
4. It's important that you **think aloud**
   please say what you're doing, what you're looking for, and what you're feeling.
5. I'll be here to take notes, but please ignore me and focus on the task.
6. We will record this session to help with analysis, and your data will remain confidential.
7. Please read and sign this consent form before we begin.

Pre-interview

Do you have any questions before we start?
I would like to **ask you a few questions**

1. Can you walk me through your typical daily commute?
2. What challenges do you face when using walking, cycling, or public transport?
3. What usually influences your decision to drive instead of using other options?
4. Have you used any travel or route planning apps before? What was your experience?

Tasks

I have a few **tasks** for you. Do the tasks as well as you can and **please think aloud the whole time.**

The first task for you:

Using the Travelly app (Google maps, Trans Link or any other app that you usually use):

- Find the **safest route**
- Explore any features that help you **understand safety or comfort**
- Start or simulate tracking your trip

While doing this, tell me:

- What you are looking for
- What confuses you

- What you like or dislike

Second task :

Now imagine you want to choose
a **healthier or more environmentally friendly route** instead of the fastest one.

- Can you find a route that supports this goal?
- What features help or don't help you decide?

Thank you. That completes the tasks.
I'd like to ask you a few follow-up questions:
Survey/interview questions

1. What 3 things did you like about the app?
2. What 3 things would you improve?
3. Did you feel confident using the app? Why or why not?
4. Would this app influence you to change your travel behaviour?

Thank you for your support.
Stop recording and start analysis
Create 3 mind maps

1. What worked well?
2. What problems and questions emerged?
3. How can the website be improved?

10.3 Appendix C: Anonymized Study Data.

All anonymized study data is placed in OneDrive folder, access is granted. Can be accessed using the link below

https://connectqutedu-my.sharepoint.com/:f:/r/personal/n12448621_qut_edu_au/Documents/Shared/IFN637-Human-Centred%20Design%20of%20IT%20Systems/Assignment-2/User%20Research%20Transcripts?csf=1&web=1&e=nQsSQJ

10.4 Appendix D: Recordings

All the recording are in OneDrive folder, access is granted. Can be accessed using the link below

https://connectqutedu-my.sharepoint.com/:f:/r/personal/n12448621_qut_edu_au/Documents/Shared/IFN637-Human-Centred%20Design%20of%20IT%20Systems/Assignment-2/User%20Research%20Recordings?csf=1&web=1&e=PDccgt